PV101 Day 3

2026-07-08 | Tim with trail crew

Rich leads today

Morning: Chapters 3-4 reading quiz

Afternoon: wire stripping and crimping lab

6 students: 2 groups of 3

Today's Target

Understand the solar system path

Use PV vocabulary without guessing

Prepare wire cleanly

Make and inspect crimped practice connections

Slow is smooth. Smooth becomes reliable.

Morning Quiz

Based on Solar Photovoltaics Basics, Chapters 3 and 4

Individual first

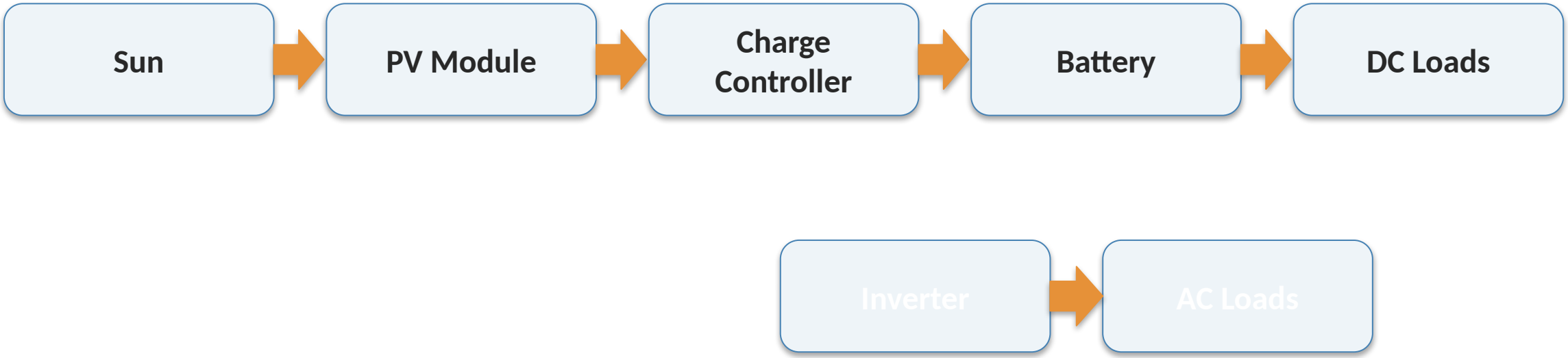
Review together after

Correct mistakes in a different color

Missed questions tell us what to reteach

PV System Path

Draw this before explaining details



Keep asking: What is the source? What stores energy? What controls it? What uses it?

Reading Frame: Chapters 3-4

Sunlight is the resource

PV cells convert light into DC electricity

Modules combine cells

Arrays combine modules

System parts control, store, and use energy

Solar Resource Words

Irradiance: sunlight power right now

Insolation: solar energy over time

Both affect how much useful energy a PV system can make

Weather, tilt, direction, and shade all matter

Power Language

Voltage = electrical pressure

Current = electrical flow

Power = work being done

Watts = Volts x Amps

Energy = power over time

Why Wire Prep Matters

Bad wiring creates unreliable systems

Loose connections become troubleshooting nightmares

Clean work is easier to inspect

Serviceability matters for the next person

Strip Quality

Good connections start before the crimp

Bad: too much copper



Good: clean strip



Bad: damaged strands



Inspect before crimping. If it looks wrong now, it will not become right later by wishing. Humans keep trying this anyway.

Crimping Rules

Match connector to wire size

Insert wire fully

Crimp on the metal barrel

Use the correct die/slot

Do not crimp loose insulation and call it optimism

Tug Test + Visual Check

Pull with reasonable hand force

Connector should not slide off

No bare copper hairs sticking out

No missing strands

Connection should be easy to inspect later

Lab Groups

Group A: 3 students

Group B: 3 students

Roles: Tool Lead, Quality Inspector, Recorder

Rotate after each completed connection

Lab Rule

Practice wire only

No live circuits

No unknown equipment

Ask Rich before changing wire, connector, or tool

Quality before speed

Exit Ticket

One thing I can do now

One thing I still need to practice

Why clean wiring matters in RV solar

Tools counted and benches cleaned before leaving